

**top
long**

**top
short**

**pant
long**

**pant
short**

~~art_four_
merged
35% 33.33%
freeze~~

~~op_four_
merged
freeze~~

**smolvla_four_
merged**

68.33% 62.5%

11.67%

38.33%

85%

What is smolvla?

Can we customize smolvla?

--task_description "fold the garment on the table"

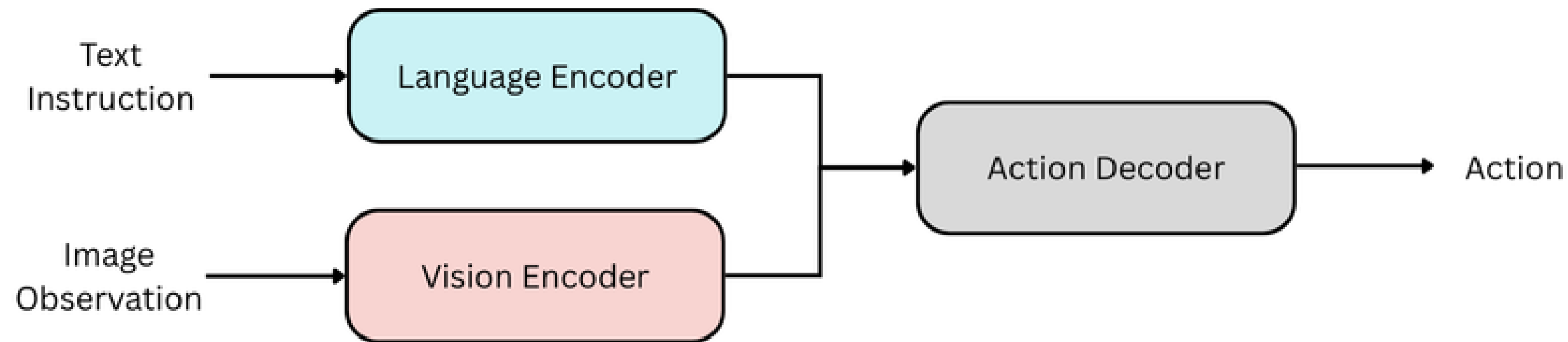
How to customize prompt?

What is smolvla?

Policy 政策	Type 類型	Description 描述	Config File 設定檔
act	Imitation Learning 模仿學習	Action Chunking Transformer 動作分塊變形金剛	configs/train_act.yaml
diffusion	Imitation Learning 模仿學習	Diffusion Policy 擴散政策	configs/train_dp.yaml
smolvla	Vision-Language-Action 視覺-語言-行動	Small Vision-Language-Action Model 小型視覺-語言-行動模型	configs/train_smolvla.yaml

In robot learning, a **vision-language-action model (VLA)** is a class of multimodal foundation models that integrates vision, language and actions. Given an input image (or video) of the robot's surroundings and a text instruction, a VLA directly outputs low-level robot actions that can be executed to accomplish the requested task.^[1]

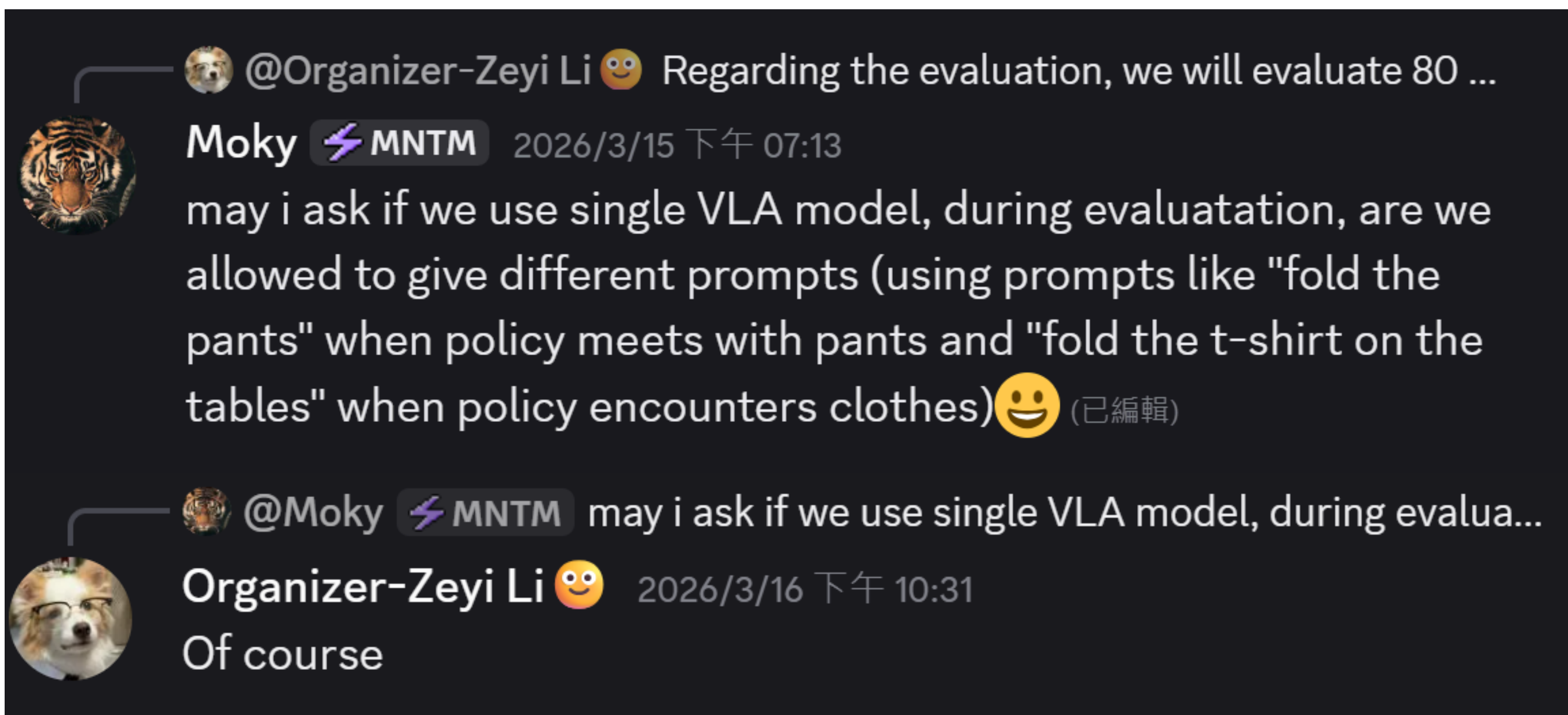
在機器人學習中，**視覺-語言-動作模型（VLA）**是一類多模態基礎模型，它整合了視覺、語言和動作。給定機器人周圍環境的輸入影像（或影片）和文字指令，VLA可以直接輸出可用於完成所請求任務的底層機器人動作。^[1]



VLA is generally constructed by fine-tuning a vision-language model (VLM) (i.e. a large language model extended with vision capabilities) on a large-scale dataset that pairs visual observation and language instructions with robot trajectories.^[2] These models combine a vision-language encoder (vision transformer), which translates an image observation and a natural language description into a distribution within a latent space, with an action decoder that transforms this representation into continuous output actions, directly executable on the robot.^[3]

視覺語言架構 (VLA) 通常透過在大規模資料集上微調視覺語言模型 (VLM) (即擴展了視覺功能的大型語言模型) 來構建，該資料集將視覺觀察和語言指令與機器人軌跡配對。^[2] 這些模型結合了視覺語言編碼器 (視覺轉換器) 和動作解碼器。視覺語言編碼器將影像觀察和自然語言描述轉換為潛在空間中的分佈，而動作解碼器則將這種表示轉換為連續的輸出動作，這些動作可以直接在機器人上執行。^[3]

Can we customize prompt?



Can we customize prompt?

Submission 提交

Once you are satisfied with your model's performance, submit your results through this [google form](#). In summary, the submitted files include:

如果您對模型的性能感到滿意，請透過此 [Google 表單](#) 提交結果。簡而言之，提交的文件包括：

- A README (.md) file that provides detailed instructions on how we can evaluate your policy.
README (.md) 文件，其中提供了有關我們如何評估您的保單的詳細說明。
- A docker image URL or huggingface repo link that contains your checkpoints, source code and any other dependencies required for evaluation.
包含檢查點、原始程式碼和評估所需的任何其他依賴項的 Docker 映像 URL 或 Hugging Face 倉庫連結。
- All the json files, one for each rollout you performed (up to 500).
所有 json 文件，每個部署對應一個文件（最多 500 個）。
- Source Code File(Optional, but highly encouraged). You can provide a link to your source code repository. This will NOT be used for evaluation, but it allows us to debug things easier if anything goes wrong during evaluation.
原始碼檔案（可選，但強烈建議提供）。您可以提供原始碼倉庫的連結。此文件不會用於評估，但如果評估過程中出現任何問題，它可以幫助我們更輕鬆地進行偵錯。

How to customize prompt?

meta/tasks.parquet



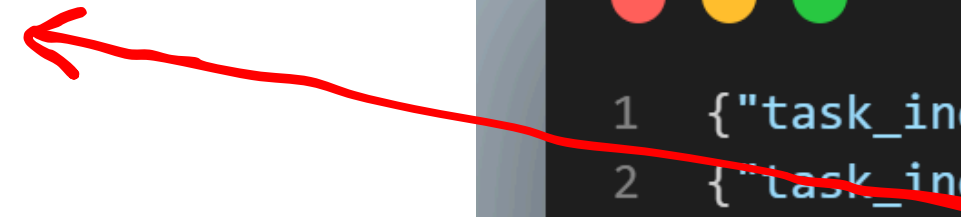
```
1 {"task_index":0,"__index_level_0__":"fold the garment on the table"}
```



```
1 {"task_index":0,"__index_level_0__":"use both arms to smooth out and fold the long pants on the table"}
2 {"task_index":1,"__index_level_0__":"use both arms to smooth out and fold the short pants on the table"}
3 {"task_index":2,"__index_level_0__":"use both arms to smooth out and fold the long-sleeve top on the table"}
4 {"task_index":3,"__index_level_0__":"use both arms to smooth out and fold the short-sleeve top on the table"}
5
```

data/chunk-000/file-000.parquet

```
{  
  "observation.state" : [...],  
  "action" : [...],  
  :  
  "task_index" : 0  
}
```



```
1 {"task_index":0,"__index_level_0__":"use both arms to smo  
2 {"task_index":1,"__index_level_0__":"use both arms to smo  
3 {"task_index":2,"__index_level_0__":"use both arms to smo  
4 {"task_index":3,"__index_level_0__":"use both arms to smo  
5
```

meta/info.json

```
{
  "codebase_version": "v3.0",
  "robot_type": null,
  "total_episodes": 1000,
  "total_frames": 265798,
  ✨ "total_tasks": 4,
  "chunks_size": 1000,
  "data_files_size_in_mb": 100,
  "video_files_size_in_mb": 200,
  "fps": 30,
  "splits": {
    "train": "0:1000"
  },
  "data_path": "data/chunk-{chunk_index:03d}/file-{file_index:03d}",
  "video_path": "videos/{video_key}/chunk-{chunk_index:03d}/video-{video_index:03d}.mp4",
  "features": {
    "observation.state": {
      "dtype": "float32",
      "shape": [
        12
      ],
    },
  },
}
```

does it work?

```
Total Episodes: 60
Average Return: 190.89 ± 64.80
Success Rate: 11.67%
```

Per-Garment Summary

```
Top_Short_Seen_0: Success Rate = 20.00%, Avg Return = 193.82
Top_Short_Seen_1: Success Rate = 0.00%, Avg Return = 152.65
Top_Short_Seen_2: Success Rate = 60.00%, Avg Return = 172.80
Top_Short_Seen_3: Success Rate = 0.00%, Avg Return = 274.65
Top_Short_Seen_4: Success Rate = 0.00%, Avg Return = 277.32
Top_Short_Seen_5: Success Rate = 60.00%, Avg Return = 161.72
Top_Short_Seen_6: Success Rate = 0.00%, Avg Return = 190.66
Top_Short_Seen_7: Success Rate = 0.00%, Avg Return = 179.64
Top_Short_Seen_8: Success Rate = 0.00%, Avg Return = 201.33
Top_Short_Seen_9: Success Rate = 0.00%, Avg Return = 150.26
Top_Short_Unseen_0: Success Rate = 0.00%, Avg Return = 159.64
Top_Short_Unseen_1: Success Rate = 0.00%, Avg Return = 176.13
```

```
Evaluation completed successfully
```

```
Total Episodes: 60
Average Return: 128.04 ± 49.29
Success Rate: 1.67%
```

Per-Garment Summary

```
Top_Short_Seen_0: Success Rate = 0.00%, Avg Return = 107.02
Top_Short_Seen_1: Success Rate = 0.00%, Avg Return = 112.51
Top_Short_Seen_2: Success Rate = 0.00%, Avg Return = 115.04
Top_Short_Seen_3: Success Rate = 20.00%, Avg Return = 192.03
Top_Short_Seen_4: Success Rate = 0.00%, Avg Return = 220.01
Top_Short_Seen_5: Success Rate = 0.00%, Avg Return = 81.73
Top_Short_Seen_6: Success Rate = 0.00%, Avg Return = 96.76
Top_Short_Seen_7: Success Rate = 0.00%, Avg Return = 103.97
Top_Short_Seen_8: Success Rate = 0.00%, Avg Return = 107.53
Top_Short_Seen_9: Success Rate = 0.00%, Avg Return = 105.13
Top_Short_Unseen_0: Success Rate = 0.00%, Avg Return = 108.43
Top_Short_Unseen_1: Success Rate = 0.00%, Avg Return = 186.38
```

```
Evaluation completed successfully
```

dataset_replay.py

data collection

```
xvfb-run -a python -m scripts.dataset_sim replay \  
--headless \  
--dataset_root Datasets/example/four_types_merged \  
--output_root Datasets/replayed \  
--save_successful_only \  
--device cpu
```

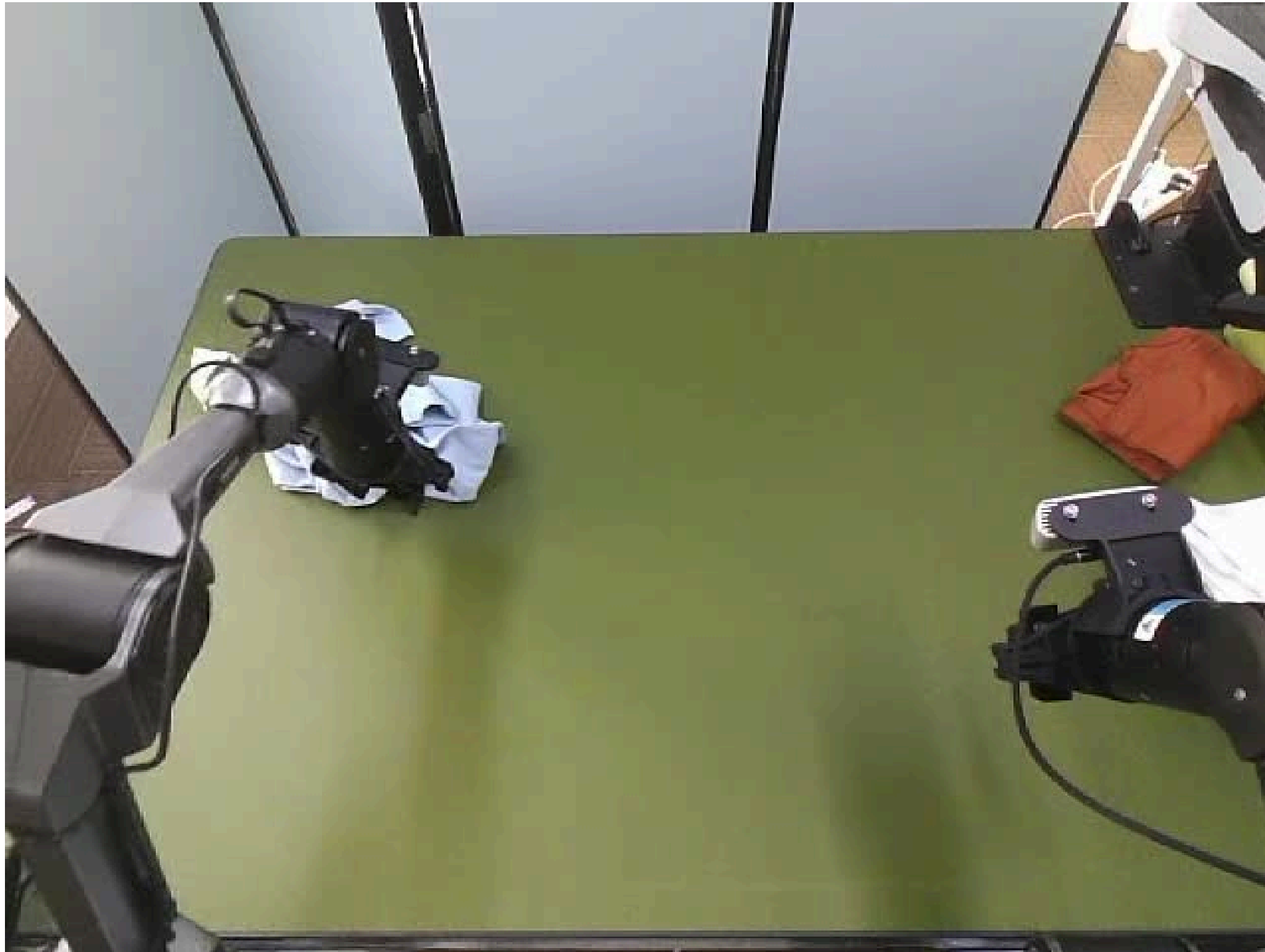

Aspect	Original Dataset	Replay Dataset
RGB Images	Rendered at recording time	Re-rendered; pixel-level differences
Depth Maps	Depth captured at recording time	Regenerated depth
Garment Shape	Garment state during manipulation	Deviates due to particle physics non-determinism
Success Rate	Manually labeled (pressing N key)	Automatically determined by <code>_check_success()</code>
Episode Count	All episodes	Only successful ones if <code>--save_successful_only</code> is used

Some update

Some update

```
✓ README.md  +7 -2  <>  ..
↑....  @@ -170,9 +170,14 @@ Evaluation is performed on the `Release` set of garments. Under the directory `A
170 170
171 171  ## 🚩 Submission
172 172
173 - Once you are satisfied with your model's performance, follow these steps to submit your results to the competition leaderboard:
173 + Once you are satisfied with your model's performance, submit your results through this [google form]
    (https://docs.google.com/forms/d/e/1FAIpQLSeeFpV4oYxSizNCnplX1Ew--TBafIbVyFg9NPH-hunks2rc7Q/viewform). In summary, the submitted files include:
174 174
175 - >Submission instructions will be available on the [competition website](https://lehome-challenge.com/).
175 + - A **README (.md) file** that provides detailed instructions on how we can evaluate your policy.
176 + - A **docker image URL** or **huggingface repo link** that contains your checkpoints, source code and any other dependencies required for evaluation.
177 + - All the json files, one for each rollout you performed (up to 500).
178 + - Source Code File(Optional, but highly encouraged). You can provide a link to your source code repository. This will NOT be used for evaluation, but it
    allows us to debug things easier if anything goes wrong during evaluation.
179 +
180 + >Please check the google form for more detailed submission information.
176 181
177 182  ## 🌸 Acknowledgments
178 183
.....
↓
```

Something just for fun



https://huggingface.co/datasets/lerobot/xvla-soft-fold/blob/main/videos/observation.images.cam_high/chunk-000/file-000.mp4


```
"names": [
    "eef_euler_0",
    "eef_euler_1",
    "eef_euler_2",
    "eef_euler_3",
    "eef_euler_4",
    "eef_euler_5",
    "eef_euler_6",
    "eef_euler_7",
    "eef_euler_8",
    "eef_euler_9",
    "eef_euler_10",
    "eef_euler_11",
    "eef_euler_12",
    "eef_euler_13",
    "eef_quat_0",
    "eef_quat_1",
    "eef_quat_2",
    "eef_quat_3",
    "eef_quat_4",
    "eef_quat_5",
    "eef_quat_6",
    "eef_quat_7",
    "eef_quat_8",
    "eef_quat_9",
    "eef_quat_10",
    "eef_quat_11",
    "eef_quat_12",
    "eef_quat_13",
    "eef_quat_14",
    "eef_quat_15",
    "eef6d_0",
    "eef6d_1",
    "eef6d_2",
    "eef6d_3",
    "eef6d_4",
    "eef6d_5",
    "eef6d_6",
    "eef6d_7",
    "eef6d_8",
    "eef6d_9",
    "eef6d_10",
    "eef6d_11",
    "eef6d_12",
    "eef6d_13",
    "eef6d_14",
    "eef6d_15",
    "eef6d_16",
    "eef6d_17",
    "eef6d_18",
    "eef6d_19",
    "eef_left_time",
    "eef_right_time",
    "qpos_0",
    "qpos_1",
    "qpos_2",
    "qpos_3",
    "qpos_4",
    "qpos_5",
    "qpos_6",
    "qpos_7",
    "qpos_8",
    "qpos_9",
    "qpos_10",
    "qpos_11",
    "qpos_12",
    "qpos_13",
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    "qvel_1",
    "qvel_2",
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    "qvel_4",
    "qvel_5",
    "qvel_6",
    "qvel_7",
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    "qvel_9",
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    "effort_4",
    "effort_5",
    "effort_6",
    "effort_7",
    "effort_8",
    "effort_9",
    "effort_10",
    "effort_11",
    "effort_12",
    "effort_13",
    "qpos_left_time",
    "qpos_right_time"
]
```

```
"names": [
    "left_shoulder_pan",
    "left_shoulder_lift",
    "left_elbow_flex",
    "left_wrist_flex",
    "left_wrist_roll",
    "left_gripper",
    "right_shoulder_pan",
    "right_shoulder_lift",
    "right_elbow_flex",
    "right_wrist_flex",
    "right_wrist_roll",
    "right_gripper"
]
```

